

Position-decoding deviations covary between hippocampal CA1 and retrosplenial cortex

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Abstract

Different brain regions can represent where an animal is, but it is less clear whether their estimates drift together from moment to moment. Such coordination has been reported between CA1 and visual or prefrontal cortex, but had not been tested between CA1 and retrosplenial cortex (RSC). We reanalyzed simultaneous CA3, CA1, and deep-layer RSC recordings from freely moving mice. Separate decoders estimated position from each region. We then asked a simple question: within the same traversal, did CA1 and RSC tend to place the animal ahead or behind its measured position at the same time? We accounted for position, nonlinear effects of running speed, and progress through the session, and compared the observed relation with 9,999 trial rotations that broke same-traversal alignment. In 12 exploratory sessions, the equal-mouse excess Fisher correlation was 0.2042 ($p = 0.0001$; four of four mice positive). A prespecified one-shot analysis of three reserved sessions confirmed the effect (excess Fisher $z = 0.1465$, $p = 0.0001$; three of three mice positive). The result remained after accounting for overall firing across CA1, RSC, and CA3 (0.1331), for the independently decoded CA3 deviation (0.1392), and for the choice of tracking frames (0.1524; all $p = 0.0001$). Thus, CA1 and the source-defined non-narrow RSC population showed coordinated short-timescale deviations during navigation. The data do not identify subjective position, a causal direction, or replication in new animals.

1 Introduction

The hippocampus carries a population code for location [1, 2], and that code can be used to estimate an animal’s position from neural activity [3]. Yet navigation is not supported by the hippocampus alone. Retrosplenial cortex (RSC) represents position, route progress, goals, landmarks, boundaries, and heading in forms that partly overlap with, but are not identical to, hippocampal coding [7, 8, 10–14]. Its strong anatomical and functional links with the hippocampal system make RSC a plausible partner in maintaining an online estimate of location [9, 15, 16].

Previous work shows that independently decoded spatial errors can move together across brain regions. This has

been observed between CA1 and visual cortex, between CA1 and prefrontal cortex, and in the joint activity of hippocampal and visual neurons [4–6]. Related coordination has also been found for decoded heading in anterior thalamus and RSC [18]. The open question here is more specific: do CA1 and RSC share short-timescale deviations in their estimates of position?

The source dataset contains simultaneous CA3, CA1, and deep-layer RSC recordings during linear-track navigation. The original study found structured patterns of communication and timing relationships along the CA3–CA1–RSC axis [17]. It did not test whether separately decoded CA1 and RSC positions shift in the same direction on individual traversals. Two regions can both represent position well without sharing these moment-to-moment shifts, so this question is not answered by spatial tuning or population coupling alone.

We therefore asked whether an RSC decoder tended to place the animal ahead when a CA1 decoder placed it ahead, and behind when CA1 placed it behind, after accounting for measured position and available behavior. We developed the analysis in 12 single-maze sessions. Before opening three reserved fixed-condition sessions, we fixed the estimator, null model, calibration, provenance, and decision rule.

2 Results

2.1 CA1 and RSC estimates drift together within a traversal

Separate CA1 and RSC decoders were trained for each direction and maze block. A decoder trained on odd traversals predicted even traversals, and the reverse decoder predicted odd traversals. Each decoder conditioned on the total number of spikes, so position was inferred from how spikes were distributed across cells rather than from overall firing alone. We retained every locomotor window, including windows with no spikes. For each region, we removed systematic effects of exact position within a bin, nonlinear running speed, and trial order. We then averaged the remaining signed deviations within each traversal and position bin.

The primary measure was the correlation between CA1 and RSC deviations in each block, direction, and held-out

fold. We transformed the correlations to Fisher z values and gave equal weight to folds, directions, blocks, sessions within each mouse, and finally mice. To ask whether same-traversal alignment mattered, we repeatedly rotated the RSC traversal labels while leaving each traversal’s spatial pattern intact. The 9,999 rotations included every allowed shift, including zero, and were averaged through the same hierarchy as the observed data.

In discovery, the observed CA1–RSC correlation exceeded the rotation reference by Fisher $z = 0.2042$ ($p = 0.0001$). The difference was positive in all four mice and in 11 of 12 sessions. The only negative session, M02-20240312, contained seven active source-defined RSC units.

2.2 The prespecified held-out test confirmed the result

The held-out test used exactly three fixed-condition sessions, one each from M01, M02, and M03, with two maze blocks per session. These recordings were not used in discovery, although the same animals were represented there. All six blocks met the prespecified requirements for cell counts and traversals.

The observed held-out value was $z = 0.1660$, whereas the mean rotation value was 0.0195. Their difference was 0.1465 ($p = 0.0001$; Fig. 1). The effect was positive in all three mice, after leaving out each mouse in turn, in both running directions, in both training/test folds, and in all six maze blocks. Nineteen of the 24 block-by-direction-by-fold records were positive.

2.3 Overall firing, CA3, and tracking noise do not explain the result

We first added the total number of spikes in CA1, RSC, and CA3 to the adjustment model. The held-out difference decreased modestly to 0.1331 but remained larger than all 9,999 rotated values ($p = 0.0001$; three of three mice positive). The relation therefore was not explained solely by broad increases and decreases in firing, or by both regions being silent at the same time.

We next accounted for the independently decoded CA3 deviation in both CA1 and RSC. The held-out difference remained 0.1392 ($p = 0.0001$; three of three mice positive). This analysis does not identify a pathway. It shows only that a single, aligned CA3 position deviation did not fully explain the CA1–RSC relation.

There is a simple tracking artifact to guard against: subtracting the same noisy position measurement from two decoded estimates can make their raw errors look correlated. After the fixed position adjustment used here, errors defined from continuous position and from the center of each spatial bin were numerically equivalent; their largest difference was 3.81×10^{-12} . Rebuilding the animal’s trajectory from the opposite half of the tracking frames also preserved the held-out result (0.1524, $p = 0.0001$; three of three mice positive). These checks make frame-by-frame

tracking noise an unlikely explanation.

3 Discussion

The central finding is straightforward: within the same traversal and part of the track, CA1 and deep-layer RSC decoders tended to be ahead or behind together. The relation remained after accounting for measured position, nonlinear speed, progress through the session, overall firing, and the decoded CA3 deviation. It was also present in sessions that were not used to choose the analysis. This extends earlier observations of coordinated spatial estimates in CA1–visual and CA1–prefrontal networks [4, 5] to the hippocampal–retrosplenial axis.

The result does not mean that CA1 sends an error signal to RSC. The shared deviations could arise through direct or indirect coupling, common inputs, an unmeasured aspect of behavior, or a broader network fluctuation. Only position and speed were available in the released behavioral data. Unlike the CA1–visual study [4], this dataset contained no independent licking or choice-error measure that could reveal where the animal believed it was. We therefore describe a “decoded-position deviation,” not “subjective position.”

The held-out sessions protect against choosing an analysis that fits only the exploratory recordings, but they are not a new-animal replication: M01, M02, and M03 were already present in discovery. The population claim is therefore limited to the four recorded mice. The source-defined RSC population also excluded narrow interneurons but included wide-spiking cells. A stricter analysis using only units labeled “Pyramidal Cell” could not be completed in every held-out block because M01 had four eligible labeled cells per block, below the prespecified minimum of five. The clearest next experiment is a replication in new animals, with histologically identified RSC cell classes and an independent behavioral report of navigation error.

The CA3 analysis suggests a second next step. Time-lagged models could test whether a shared position deviation appears first in one region and later in another, without treating simultaneous correlation as evidence of direction. That study should be specified before examining new data and should keep two ideas separate: a shared neural estimate of position and a subjective map validated by behavior.

4 Methods

4.1 Dataset and separation of discovery from confirmation

We analyzed the navigation recordings released with Gonzalez et al. [17] (DANDI:001695, version 0.260319.2023). Discovery comprised 12 single-maze sessions: M01-20240312/13/14, M02-20240312/13/14, M03-20240621/22/23, and M05-20240729/30/31. Three fixed-condition sessions were sealed for confirmation: M01-20240318, M02-20240318, and M03-20240624. Each con-

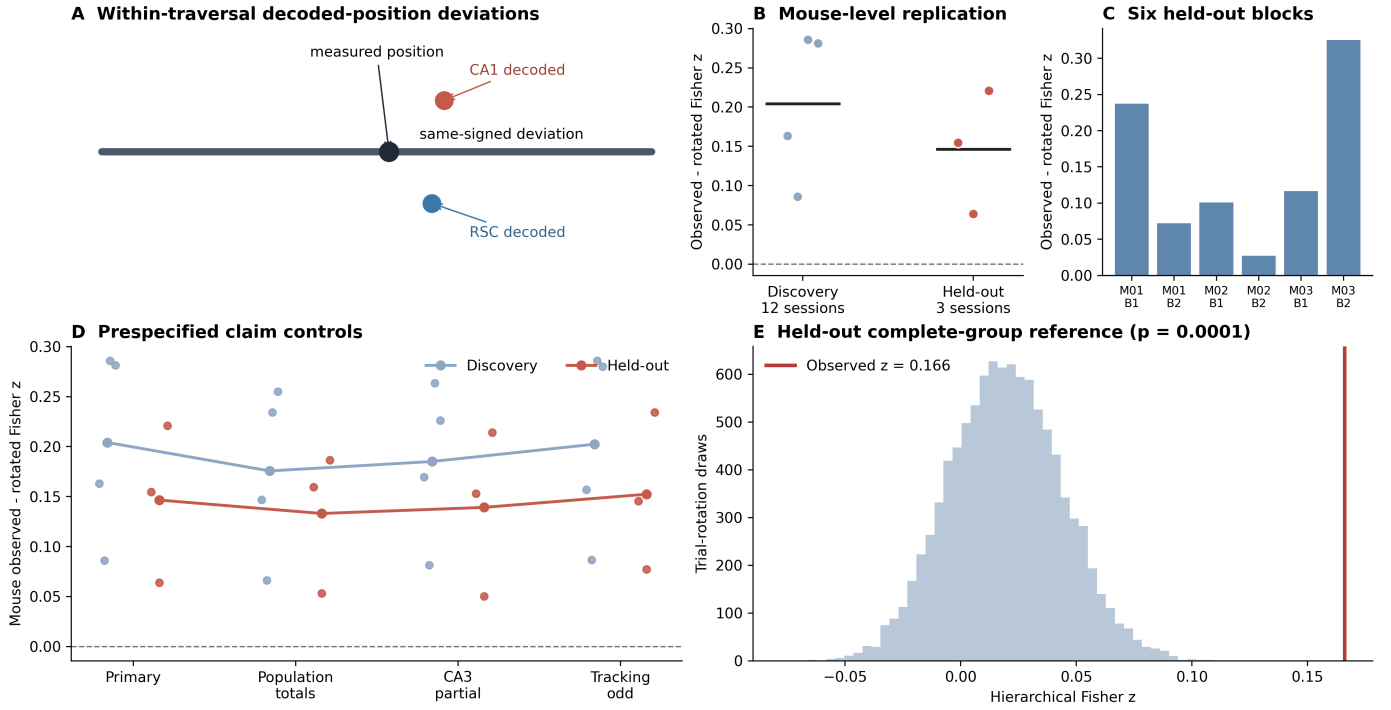


Figure 1: CA1 and RSC position estimates deviate together in held-out sessions. **A**, Core idea: two independently decoded positions fall ahead of or behind measured position together. The analysis uses deviations after accounting for measured behavior. **B**, Observed-minus-rotation Fisher correlations for each mouse in exploratory and held-out sessions; horizontal bars show cohort means. **C**, The difference was positive in all six held-out maze blocks. **D**, It remained positive after accounting for overall firing in the three regions, decoded CA3 deviation, and the choice of tracking frames. Points are mice and lines are equal-mouse means. **E**, Prespecified held-out value (red) relative to 9,999 trial rotations.

firmation session contained two maze blocks. Before confirmation, code, simulation calibration, input hashes, seed, draw counts, and the complete decision rule were committed publicly. A write-once sentinel was created before the first aligned held-out score.

CA1 and CA3 populations used units labeled “Pyramidal Cell.” The primary RSC population followed the source convention `cell_type != Narrow Interneuron`. Units required at least 20 locomotor spikes within a block. Blocks required at least ten CA1 and five RSC units and at least 16 accepted traversals per direction.

4.2 Behavioral coordinate and temporal observations

Position was reduced to the track’s main axis and normalized from zero to one using the first and 99th percentiles within each maze block. Traversals connected the 0.15 and 0.85 endpoints, covered at least 0.70 of the track, and lasted no more than 120 s. Within each traversal, pool-adjacent-violators regression produced a monotonic path from even-indexed tracking frames; this path was interpolated to nonoverlapping 250-ms neural windows. Windows required speed above 2.5 cm/s. The decoder used 24 fixed physical bins and was evaluated in central bins 4–19.

4.3 Conditional-multinomial decoding

For position bin x , area-specific cell probabilities were estimated from training spikes as

$$p_i(x) = \frac{K_i(x) + \alpha}{\sum_j K_j(x) + \alpha N}, \quad (1)$$

where $K_i(x)$ is the one-bin Gaussian-smoothed spike count for cell i , N is the number of included cells, and $\alpha = 0.5$. For an observed count vector $\mathbf{k}$ with total n , the position likelihood conditioned on n was

$$\log L(x | \mathbf{k}, n) = \sum_i k_i \log p_i(x) + C(\mathbf{k}), \quad (2)$$

with a uniform spatial prior. The decoded position was the posterior mean of the 24 bin centers. When $n = 0$, the posterior was uniform and the prediction was 0.5; these windows were retained.

Decoders were fitted separately for each brain region, maze block, running direction, and traversal parity. Even traversals predicted odd traversals and vice versa, so no test traversal contributed spikes to the decoder used on it.

4.4 Covariate adjustment and record statistic

At each held-out 250-ms window, signed error was decoded position minus measured position. In each block-by-direction-by-fold record, we separately adjusted CA1,

RSC, and CA3 errors for 16 position-bin indicators, linear, quadratic, and cubic terms for position within each bin, linear, quadratic, and cubic standardized speed, and linear and quadratic traversal order. The remaining deviations were averaged within traversal-by-position cells. The record-level measure was the Pearson correlation between CA1 and RSC deviations, transformed to Fisher z before hierarchical averaging.

The population-total control added log-transformed CA1, RSC, and CA3 spike totals from each 250-ms window to the residual design. For the CA3 control, a cubic function of the aggregated decoded CA3 residual was removed from both CA1 and RSC before their correlation.

4.5 Trial-rotation inference

For every draw and block-by-direction-by-fold group, RSC traversal rows were cyclically shifted by an independently sampled integer from zero through $n - 1$. The full RSC row, its position columns, and its missing-bin mask moved together. This preserved each traversal’s spatial pattern while breaking its alignment with the matching CA1 traversal. All 9,999 draws were averaged using the same equal weighting of folds, directions, blocks, sessions within mouse, and mice as the observed value. One-sided Monte Carlo values used

$$p = \frac{1 + \#\{T_{\text{null}} \geq T_{\text{obs}}\}}{1 + 9999}. \quad (3)$$

Before confirmation, 1,000 synthetic null cohorts produced a 5.0% rejection rate, and 1,000 cohorts with correlation 0.10 produced 99.9% power using 199-draw calibration tests. The confirmation required positive aggregate excess, $p \leq 0.05$, positive margins in all three mice and all leave-one-mouse-out estimates, and positive margins for both directions and both folds.

Data and code availability

All analysis code, frozen specifications, calibration draws, discovery outputs, confirmation null banks, provenance manifests, and figures are public at <https://github.com/asim-v/ca1-rsc-coherent-position-errors>. Raw data are available from DANDI:001695.

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